

CATANDUANES RADAR, Philippines

WMO: 984470

Lat: 13.6298N

Lon: 124.3340E

Elev: 233

StdP: 98.56

Time Zone: 8.00 (E08)

Period: 94-12

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	21.2	21.8	19.0	14.2	22.2	19.6	14.8	22.6	10.2	23.5	8.5	24.0	3.8	40	0.376

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	4.0	30.5	26.9	30.0	26.7	29.6	26.5	27.5	29.6	27.1	29.3	26.8	28.9	3.9	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.8	23.1	29.1	26.5	22.6	28.8	26.2	22.2	28.5	88.9	29.8	87.1	29.5	85.9	29.2	29.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.8	8.2	7.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	25.9	24.2	24.5	25.2	26.1	26.9	27.0	26.7	26.8	26.7	26.3	25.7	24.8
	DBStd	1.41	1.03	1.04	1.13	0.90	1.06	1.12	1.05	1.00	1.07	1.04	0.99	1.09
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	5807	439	407	472	483	522	511	517	521	500	505	471	459
	CDD18.3	2766	181	174	213	233	264	261	259	262	250	247	221	201
	CDH23.3	21269	692	806	1324	1849	2516	2515	2357	2363	2183	1987	1603	1074
(14) Wind	WSAvg	3.9	4.4	4.1	3.9	3.6	3.3	3.3	3.9	4.0	4.0	3.5	4.1	4.5
	PrecAvg	3795	385	274	228	198	218	260	245	189	244	405	520	632
	PrecMax	5596	1054	1054	593	923	727	464	468	396	526	908	1126	1712
	PrecMin	2274	96	43	27	36	52	81	103	59	84	160	89	183
	PrecStd	1070	237	236	169	215	171	100	102	75	115	169	265	361
	DB	27.7	28.2	28.8	29.7	31.1	31.2	30.4	30.8	30.9	30.4	29.1	28.6	28.6
	MCWB	25.2	25.5	25.6	26.4	27.1	27.3	26.7	26.8	27.0	26.9	26.4	26.1	26.1
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	27.0	27.6	28.2	29.0	30.0	30.2	29.8	30.1	30.0	29.4	28.5	27.6	27.6
	2% MCWB	24.8	25.1	25.3	26.1	26.7	27.0	26.6	26.5	26.6	26.4	26.1	25.5	25.5
	5% DB	26.4	27.0	27.7	28.5	29.4	29.6	29.3	29.5	29.3	28.8	28.0	27.0	27.0
	5% MCWB	24.4	24.6	25.1	25.8	26.5	26.8	26.4	26.3	26.3	26.1	25.9	25.1	25.1
	10% DB	25.9	26.4	27.1	28.1	28.8	29.1	28.7	28.9	28.7	28.2	27.5	26.4	26.4
	10% MCWB	24.0	24.3	24.9	25.6	26.3	26.5	26.1	26.1	26.0	25.9	25.6	24.8	24.8
	WB	25.8	26.1	26.2	27.2	27.7	28.1	27.5	27.9	27.4	27.4	27.0	26.6	26.6
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% MCDB	27.3	27.7	28.0	28.8	30.0	30.3	29.5	29.8	29.8	29.3	28.5	28.1	28.1
	2% WB	25.2	25.4	25.7	26.6	27.1	27.4	27.0	27.0	26.9	26.7	26.5	25.8	25.8
	2% MCDB	26.5	27.0	27.5	28.3	29.3	29.7	29.2	29.3	29.3	28.6	28.0	27.1	27.1
	5% WB	24.7	25.0	25.4	26.2	26.7	27.0	26.6	26.6	26.5	26.4	26.1	25.4	25.4
	5% MCDB	26.0	26.5	27.1	28.0	28.8	29.2	28.8	28.9	28.8	28.3	27.6	26.6	26.6
	10% WB	24.3	24.6	25.1	25.8	26.5	26.6	26.2	26.2	26.2	26.0	25.7	25.0	25.0
	10% MCDB	25.4	26.0	26.7	27.6	28.5	28.7	28.3	28.5	28.3	27.9	27.2	26.1	26.1
(35) Mean Daily Temperature Range	MDBR	2.8	3.2	3.4	3.9	3.9	4.0	3.8	4.2	4.0	3.9	3.2	2.9	2.9
	5% DB MCDBR	3.7	3.9	4.1	4.5	4.8	4.7	4.7	5.1	5.1	4.8	3.9	3.8	3.8
	5% DB MCWBR	2.4	2.5	2.4	2.6	2.7	2.7	2.7	2.8	2.8	2.7	2.5	2.4	2.4
	5% WB MCDBR	3.4	3.6	3.8	4.1	4.4	4.6	4.5	4.8	4.8	4.4	3.7	3.5	3.5
	5% WB MCWBR	2.5	2.5	2.4	2.7	2.7	2.8	2.9	3.0	2.9	2.8	2.5	2.5	2.5
(40) Clear-Sky Solar Irradiance	taub	0.449	0.425	0.432	0.416	0.421	0.419	0.422	0.447	0.449	0.441	0.448	0.466	0.466
	taud	2.319	2.403	2.390	2.449	2.461	2.482	2.473	2.382	2.370	2.395	2.354	2.262	2.262
	Ebn at Noon	848	888	887	894	874	867	866	854	858	861	843	819	819
	Edh at Noon	126	120	124	117	114	110	111	123	125	120	121	131	131
(44) All-Sky Solar Radiation	RadAvg	3.85	4.87	5.44	6.22	6.08	5.55	4.92	5.12	5.06	4.58	4.25	3.63	3.63
	RadStd	0.46	0.58	0.57	0.40	0.46	0.38	0.47	0.29	0.44	0.28	0.49	0.41	0.41

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (5 neighbors)	+0.18	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page